第一章 前言

1. **What is the main advantage of multiprogramming?**

Multiprogramming maximizes CPU utilization through concurrency.

With multiprogramming, CPU can switch to another process when one process is blocked (like waiting I/O or sth).

2. **Define the essential properties of the following types of operating systems:**

   a. Batch **b. Time sharing**
   c. Real time **d. Network**
   e. Distributed

a: executes jobs in batches without user interaction during processing.

b: shares CPU time among users to allow interactive use.

c: ensures tasks are completed within strict timing deadlines.

d: provides services and resource sharing across connected computers.

e: makes multiple computers work together as a single system.

3. 下面哪些指令是特权指令？解释原因。

   A. 设置定时器的值
   B. 读时钟
   C. 清除内存
   D. 关闭中断
   E. 从用户模式切换到监督模式

A.C.D.E是

A、C、D、E的操作都需要在kernel mode下完成，若用户操作会破坏系统正常运行。

B不会对系统产生影响。